ResearchOS Full Model Architecture

Evidence-first, multi-agent scientific research and academic writing platform

EXPERIENCE PLANE

Researcher-facing Next.js interface

Researcher
Browser / desktop
Authenticated tenant

Next.js 15 Workspace
Projects, navigation, responsive shell
React 19 + TypeScript

Academic Editor
Sections, tables, images, equations
DOCX / PDF export

AI Research Chat
Streaming answers and agents
Files, citations, run status

Library & Papers
Imported PDFs and summaries
Search and source preview

APPLICATION PLANE

FastAPI service
Tenant boundaries
REST + SSE

API Gateway
FastAPI routers + validation
CORS, readiness, OpenAPI

Identity & Tenancy
Registration, login, refresh
Organizations, authorization

Project Services
Projects and conversations
Messages, drafts, context

Source Services
Upload, library, download
Scholarly search, storage

Chat Streaming
History-aware model context
Citations, SSE, usage

Jobs API
Durable jobs
Retry, cancel, progress

RESEARCH INTELLIGENCE

Evidence pipeline
Agent orchestration
Verified outputs

EVIDENCE AND WRITING PIPELINE

Document Intelligence
PDF, tables, figures, datasets
Structured provenance

Hybrid Retrieval
Chunks, vectors, lexical fusion
Reranking

Evidence & Verification
Evidence packs and claim graph
Support, conflict, citations

Academic Writing
Contextual continuation
IEEE / APA / MLA references

Model Gateway
Provider-neutral streaming
OpenAI and compatible APIs
Gemini, Ollama
Deterministic local provider

MULTI-AGENT RESEARCH RUNTIME

Context Compiler
History + evidence + memory

Planner & Scheduler
LangGraph runs and retries

Specialized Research Agents
Literature, gaps, hypotheses, novelty, feasibility
Critic, statistics, tournament, code

Tool Gateway
Permissions, MCP, integrations
Sandboxed execution

EXECUTION AND DATA

Docker Compose
Durable state
Background work

Background Worker
Document parse jobs
Durable tasks, retry, cancel

PostgreSQL
Users, projects, messages
Files, jobs, claims, runs

Redis
Coordination and queues
Runtime checkpoints

Chroma
Tenant-scoped vectors
Semantic retrieval

MinIO Object Storage
PDFs and source documents
Project-scoped object keys

EXTERNAL SYSTEMS

Controlled adapters

LLM Providers
OpenAI, compatible, Gemini, Ollama

Scholarly Providers
Metadata, open-access PDFs, summaries

MCP and Research Tools
Typed, permissioned, audited execution

Local Research Sources
PDF, CSV, JSON, text, images, library items